Charge Transport in Semiconductors

3.1 Drift current

When a small electric field is applied across a bar of semiconductor, electrons and holes acquire a drift velocity (v_d) proportional to the magnitude of the electric field. While electrons move in a direction opposite to that of the applied field, holes move in the direction of the field. This directional movement of the charge carriers constitutes a current, which is usually referred to as the drift current. It has already been derived earlier that the current density (J) in a semiconductor sample having 'n' electrons per unit volume traveling at a velocity 'v' is given by (refer to eqn.(1.5))

$$J = qnv \ldots\ldots (3.1)$$

Considering only free electrons, so that 'n' is now the free electron concentration, and replacing 'v' with the drift velocity 'v_d', the electron drift current density can be expressed as

$$J_{ndr} = -qnv_d \ldots\ldots. (3.2)$$

The minus sign in eqn. (3.2) indicates that the direction of the electron drift current J_{ndr} is opposite to that the drift velocity. From eqn. (1.66), we know that the drift velocity is proportional to the electric field (ε) for moderate electric fields and for electrons it can be expressed as $v_d = -\mu_n\varepsilon$. Substituting eqn.(1.66) in eqn. (3.2), we get

$$J_{ndr} = qn\mu_n\varepsilon \ldots\ldots\ldots\ldots\ldots\ldots (3.3a)$$

Following a similar analysis for holes, the drift hole current density can be expressed as

$$J_{pdr} = qpv_d = qp\mu_p\varepsilon \ldots\ldots\ldots\ldots\ldots (3.3b)$$

However, the above equations are valid only for moderate electric fields. For high fields, as already pointed out in Section1.5.3, the mobility is no longer a constant and $\mu\varepsilon$ has to be replaced with v_{sat}.

If the semiconductor contains both types of carriers, the total drift current density (J_{dr}) is the sum of the electron and hole current densities. Since electrons and holes move in directions opposite to each other, the total current actually adds up, and so

$$J_{dr} = qn\mu_n\varepsilon + qp\mu_p\varepsilon = \sigma\varepsilon \quad \ldots\ldots\ldots\ldots\ldots\ldots(3.4)$$

where σ is the conductivity of the semiconductor sample given by

$$\sigma = qn\mu_n + qp\mu_p \quad \ldots\ldots\ldots(3.5)$$

The resistivity (ρ), which is the reciprocal of conductivity, can therefore be expressed as

$$\rho = \frac{1}{qn\mu_n + qp\mu_p} \quad \ldots\ldots\ldots(3.6)$$

We know that the majority carrier concentration in a strongly extrinsic semiconductor is many orders of magnitude higher than the minority carrier concentration. Therefore from eqns.(3.4),(3.5) and (3.6), we can write

$$\left.\begin{aligned} J_{dr} &\approx J_{ndr} = qn\mu_n\varepsilon \\ \sigma &\approx \sigma_n = qn\mu_n \\ \rho &\approx \rho_n = \frac{1}{qn\mu_n} \end{aligned}\right\} \ldots\ldots\text{for n} - \text{type} \ldots\ldots\ldots\ldots(3.7)$$

$$\left.\begin{aligned} J_{dr} &\approx J_{pdr} = qp\mu_p\varepsilon \\ \sigma &\approx \sigma_p = qp\mu_p \\ \rho &\approx \rho_p = \frac{1}{qp\mu_p} \end{aligned}\right\} \ldots\ldots\text{for p} - \text{type} \ldots\ldots\ldots\ldots(3.8)$$

Q.3.1 : Show that the maximum resistivity for silicon is obtained when the sample is slightly p-type.

From eqn. (3.5), we see that at thermal equilibrium, the conductivity of a semiconductor sample where the electron and hole concentrations are comparable to each other is given by

$$\sigma = qn\mu_n + qp\mu_p = q(\frac{n_i^2}{p}\mu_n + p\mu_p)$$

For maximum resistivity, i.e. minimum conductivity, $\frac{d\sigma}{dp} = 0$. Differentiating the above expression with respect to the hole concentration and equating it to zero, we obtain $\frac{n_i^2}{p^2}\mu_n = \mu_p$, i.e. $p = n_i\sqrt{\frac{\mu_n}{\mu_p}}$. Since $\mu_n > \mu_p$, $p > n_i$, i.e. the sample is slightly p-type.

Example 3.1: (a) Calculate the resistivity of intrinsic silicon at room temperature;

(b) Calculate the resistivity of an n-type semiconductor with a doping concentration $N_D = 10^{16}/cm^3$ at room temperature.

(a) From Table 1.3, we know that the intrinsic carrier concentration for silicon at room temperature is $1.5x10^{10}/cm^3$. Also, from Table 1.4, the electron mobility for intrinsic silicon is $1350cm^2/V.sec$ and the hole mobilty is $480cm^2/V.sec$. The resistivity of intrinsic silicon is therefore calculated from eqn.(3.6) as

$$\rho = \frac{1}{qn_i(\mu_n + \mu_p)} = \frac{1}{1.6x10^{-19}\,x1.5x10^{10}(1350+480)} = 2.3x10^5\ ohm-cm$$

(b) Assuming complete ionization, in this case the electron concentration $n \approx N_D = 10^{16}/cm^3$. The hole concentration $p = n_i^2/N_D = 2.25x10^4/cm^3$. Since n>>p, the resistivity

of the sample is effectively given by eqn.(3.7). Assuming the electron mobility in this sample is equal to that in an intrinsic silicon sample, the resistivity is calculated to be

$$\rho = \frac{1}{qn\mu_n} = \frac{1}{1.6x10^{-19}\,x10^{16}\,x1350} = 0.462ohm - cm$$

The above example highlights the fact that by introducing a small amount of impurity (<1ppm), it is possible to change the resistivity of silicon by many orders of magnitude. As mentioned in Chapter1, this is an important property of semiconductors. Evidently, then, accurate measurement of doping concentration in a semiconductor is essential to determine its basic properties. Please note that the electron (and hole) mobility is also dependent on doping and is reduced considerably for high doping concentration, although that was not taken into account in Example 3.1. Therefore, in practice, a straightforward current-voltage measurement does not yield an accurate value of carrier concentration unless the value of electron (and hole mobility) is also known. In the next section, we shall discuss a simple way of measuring carrier concentration and mobility simultaneously in an extrinsic semiconductor.

3.2 Hall Effect

The Hall effect is widely used for directly measuring carrier concentration. The underlying principle in this measurement is that when a magnetic field is applied in a direction perpendicular to the flow of charge carriers, the path of the carriers gets deflected. Fig. 3.1 shows a p-type semiconductor bar of length 'L', width 'W' and thickness 'd'. Electric and magnetic fields are applied to this bar along x-axis and z-axis respectively. Due to the magnetic field (B_z), an upward Lorentz force given by qv_xB_z is exerted on the carriers moving along x-direction with a drift velocity v_x. This upward

force results in accumulation of holes at the top of the sample which gives rise to a corresponding downward electric field (ε_y). Since there is no net current flow in the y-direction in the steady state, the force due to electric field must exactly balance the Lorentz force, i.e. $q\varepsilon_y = qv_xB_z$, or

$$\varepsilon_y = v_xB_z \quad \text{.....(3.9)}$$

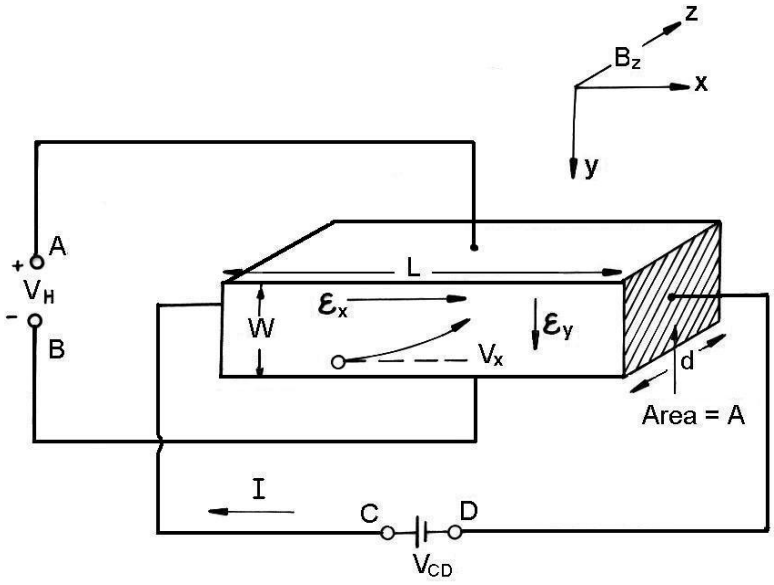


Fig.3.1: Hall Effect measurement set-up. The polarity of V_H and the direction of ε_y correspond to that for a p-type semiconductor

Physically, this means that the magnetic field results in an increase in the hole concentration at the top surface of the sample shown in Fig. 3.1, till the electric field ε_y becomes as large as v_xB_z. Once this electric field is established, no net lateral force is experienced by the holes as they drift along the bar in the x-direction. The establishment of this electric field is known as the **Hall effect** and the field itself is called the **Hall field**. The voltage across the terminals A and B is called the **Hall voltage (V_H)**. As seen from Fig. 3.1, $V_H = \varepsilon_y W$. With the help of eqns. (3.9) and (3.3b), we can now express this Hall field as

$$\varepsilon_y = v_x B_z = \frac{J_p B_z}{qp} = J_p B_z R_H \quad \text{.....(3.10)}$$

The above equation signifies that the Hall field is proportional to the product of current density and applied magnetic field with a proportionality constant $R_H = 1/qp$, called the **Hall coefficient**. A measurement of Hall voltage for a known current (I) and magnetic field is carried out to obtain the hole concentration (p) in the sample. From eqn.(3.10), we have

$$p = \frac{1}{qR_H} = \frac{J_p B_z}{q\varepsilon_y} = \frac{\left(\frac{I}{Wd}\right)B_z}{q\left(\frac{V_H}{W}\right)} = \frac{IB_z}{qV_H d} \text{.....(3.11)}$$

Since all the quantities on the right hand side are measurable, accurate values of the hole concentration can be obtained directly from Hall measurement. Now substituting $v_x = \mu_p \varepsilon_x$ in eqn.(3.10), we can write

$$\varepsilon_y = \frac{V_H}{W} = \mu_p \varepsilon_x B_z = \frac{\mu_p V_{CD} B_z}{L} \text{..... (3.12)}$$

Therefore, the mobilty can also be obtained by direct measurement as

$$\mu_p = \frac{LV_H}{WV_{CD}B_z} \text{.....(3.13)}$$

Hall measurement carried out over a wide range of temperatures yield plots of majority carrier concentration and mobility versus temperature, which are very useful data in analysis of semiconductor materials. It is evident that Hall measurement can also be carried out for n-type semiconductor and similar results with a change of sign are obtained. For an n-type semiconductor, due to the accumulation of the negatively charged electrons at the top surface of the sample as shown in Fig. 3.1, the sign of V_H and the

direction of ε_y will be reversed. Indeed, the sign of Hall voltage is a common way to determine whether the sample is n-type or p-type.

3.3 Diffusion Current

So far in our discussions, we have considered a homogeneous semiconductor material, i.e. a material with uniform doping concentration. However, if there is a spatial variation of carrier concentration in the semiconductor, then in addition to the drift current, there will exist another important current component due to this concentration gradient. When there is a concentration gradient, the carriers tend to diffuse from the region of higher concentration to the region of lower concentration. This constitutes a current flow called the **diffusion current**.

In order to derive a relation for the diffusion current, let us consider an n-type semiconductor sample where the electron concentration varies from left to right as shown in Fig. 3.2. To carry out a one-dimensional analysis, let us assume that the doping concentration varies only in the x-direction, and is constant along the cross-section. The diffusion of carriers is actually a result of their constant random motion, which we discussed in Section 1.5.1. The electrons are also constantly scattered due to collisions with lattice atoms, impurities, defects or other electrons. Let the mean free path of the electrons between collisions be l and the mean free time be τ_c so that $l = v_{th}\tau_c$, where v_{th} is the average thermal velocity. Since the sample is at uniform temperature, v_{th} is same at every point.

Let n(x) denote the concentration of electrons in the sample at any point x. Then the average concentration of electrons in the sample from x = -l to x = 0 is $\frac{n(-l)+n(0)}{2}$ and the number of electrons per unit area in the region from x = -l to x = 0 is given by

$l\left[\frac{\mathrm{n}(-l)+\mathrm{n}(0)}{2}\right]$. We assume that there is no electric field, so that the motion of electrons are perfectly random, i.e. half of them moves to the right and the other half moves to left at any given time. Therefore, the number of electrons crossing the plane at x = 0 from left in time τ_c is $\frac{l}{4}\left[\mathrm{n}(-l)+\mathrm{n}(0)\right]$.

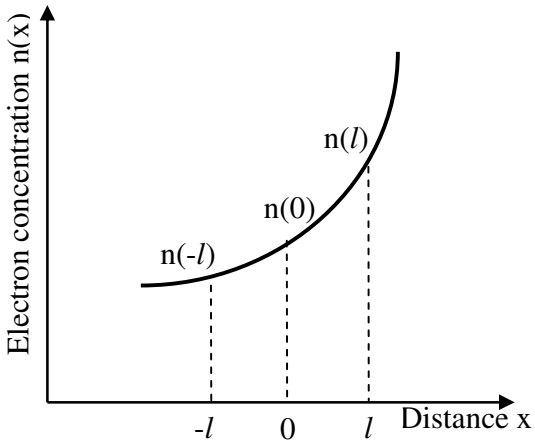


Fig.3.2: Electron concentration as a function of position showing the effect of diffusion

The current density at x = 0 is given by the net number of electrons crossing the plane at x = 0 per unit time per unit area. The flux F_1, defined as the average rate of electron flow per unit area per unit time moving from left to right, is given by

$$\mathrm{F}_1 = \frac{l}{4\tau_c}\left[\mathrm{n}(-l)+\mathrm{n}(0)\right]\ldots\ldots.(3.14)$$

Similarly, the flux F_2 defined as the average rate of electron flow per unit area per unit time moving from right to left is given by

$$\mathrm{F}_2 = \frac{l}{4\tau_c}\left[\mathrm{n}(l)+\mathrm{n}(0)\right]\ldots\ldots\ldots(3.15)$$

Therefore the net flux F can be written as

$$F = F_1 - F_2 = \frac{l}{4\tau_c}\left[n(-l) - n(l)\right] \ldots\ldots..(3.16)$$

The concentration gradient of electrons around x = 0 can be expressed as

$$\frac{dn}{dx} = -\frac{n(-l) - n(l)}{2l} \ldots\ldots\ldots(3.17)$$

From eqns. (3.16) and (3.17), we now have

$$F = -\frac{dn}{dx}\frac{l^2}{2\tau_c} = -D_n\frac{dn}{dx} \ldots\ldots.(3.18)$$

where $D_n = l^2/(2\tau_c)$ is the **diffusion coefficient** or **diffusivity** of electrons. The electron flux F constitutes an electron diffusion current density given by

$$J_{ndiff} = -qF = qD_n\frac{dn}{dx} \ldots\ldots..(3.19)$$

A similar analysis as above can be carried out considering holes resulting in a relation for the hole diffusion current density given by

$$J_{pdiff} = qF = -qD_p\frac{dp}{dx} \ldots\ldots.(3.20)$$

The negative sign in eqn. (3.20) signifies that the direction of the hole current is opposite to the direction of the increasing concentration gradient, since the movement of holes is from a point of higher concentration to a point of lower concentration. Electrons also move from a point of higher concentration to a point of lower concentration. However, since the direction of the conventional current is opposite to the direction of electron movement, J_{ndiff} is in the same direction as the increasing concentration gradient as shown in eqn. (3.19).

3.4 Current Density Equations

When an electric field as well as a concentration gradient is present across a semiconductor sample, both drift and diffusion currents will flow and the total current density will be given as the sum of the drift and diffusion components. Thus the total electron (and hole) current density is given by

$$J_n = J_{ndr} + J_{ndiff} = qn\mu_n \varepsilon + qD_n \frac{dn}{dx} \quad \text{……..(3.21a)}$$

$$J_p = J_{pdr} + J_{pdiff} = qp\mu_p \varepsilon - qD_p \frac{dp}{dx} \quad \text{……..(3.21b)}$$

An important aspect of eqn. (3.21) is that the minority carriers can contribute significantly to the total current through diffusion. As the drift current is proportional to the carrier concentration, the contribution of minority carriers is usually negligible. However, even when the actual concentration of minority carriers is very small, the gradient may be significant. Thus the minority carrier diffusion current may become as large as majority carrier currents in certain situations as we shall see later.

3.5 Einstein's Relation connecting μ and D

Although drift and diffusion are two seemingly different processes, a close relationship exists between mobility and diffusion coefficient. This is because both these parameters are determined by the thermal motion and scattering of the free carriers. In order to establish the relationship between μ and D, we shall make use of the important concept of constant Fermi Energy level at thermal equilibrium already demonstrated in Section 1.3.9.

Let us consider a bar of non-uniformly doped n-type semiconductor, with a doping concentration as shown in Fig.3.3(a), under thermal equilibrium. The bar is intrinsic for $x \leq 0$, while from x =0, the doping concentration gradually increases upto x = *l.* Beyond x = *l*, the doping becomes constant again. Let us assume complete ionization. Now, because of the concentration gradient, electrons diffuse from x = *l* towards x = 0, leaving behind positively charged ionized donors. This charge separation gives rise to an electric field, which tends to pull the electrons back towards x = *l*. Thus, in thermal equilibrium, electrons tend to diffuse down the concentration gradient causing a diffusion flow of electrons from right to left. Again, the resulting electric field causes a drift flow of electrons in the opposite direction. Since no net current flows, these two current components must add up to zero. It is easy to see why equilibrium will be established. Suppose, for some reason, the equilibrium is disturbed so that the diffusion current is more than the drift current. This will mean a net flow of electrons from right to left, resulting in an increase in positive ionized donors. Consequently, the built-in electric field and the drift of electrons from left to right increase, till equilibrium is re-established. Therefore, we can write

$$J_n = qn(x)\mu_n \varepsilon + qD_n \frac{dn}{dx} = 0 \text{ .....(3.22)}$$

Now, at x = 0, the Fermi level is at the intrinsic level, i.e. $E_i(0) = E_F$, since the semiconductor is intrinsic up to that point. Since at thermal equilibrium, Fermi level must be constant, we can write from Fig.3.3(b)

$$E_F - E_i(x) = E_i(0) - E_i(x) = q\psi(x) \text{.....(3.23)}$$

where $\psi(x)$ is the potential difference between the points 'x' and '0'. From eqn. (1.27), we can express the electron concentration variation n(x) as

$$n(x) = n_i \exp\left[\frac{E_F - E_i(x)}{kT}\right] \ldots\ldots\ldots\ldots (3.24)$$

Now, substituting eqn.(3.23) in eqn.(3.24), we can write

$$n(x) = n_i \exp\left[\frac{q\psi(x)}{kT}\right] \ldots\ldots (3.25)$$

Differentiating eqn. (3.25) with respect to x, we can write

$$\frac{dn}{dx} = \frac{q}{kT} n(x) \frac{d\psi}{dx} = -\frac{n(x)\varepsilon}{V_T} \ldots\ldots (3.26)$$

where $\mathbf{V_T = kT/q}$ is the **thermal voltage** and ε is the built-in field created due to the diffusion of electrons. Substituting eqn. (3.26) in eqn.(3.22), we have

$$D_n = \mu_n V_T \ldots\ldots (3.27a)$$

Similarly, considering holes, we can show that

$$D_p = \mu_p V_T \ldots\ldots (3.27b)$$

Equations 3.27(a) and (b) are known as **Einstein's relationship**. This important set of equations shows that there is a definite relationship between drift and diffusion.

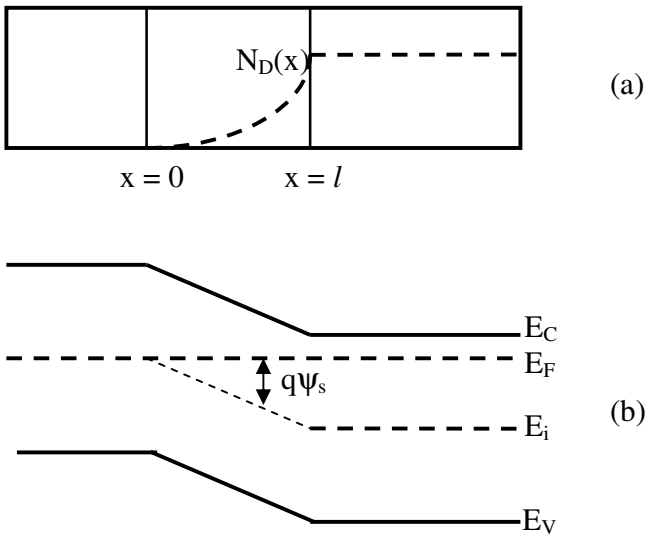


Fig. 3.3: (a) A bar of n-type semiconductor showing graded impurity distribution
(b) The energy band diagram for this bar at thermal equilibrium

3.6 Continuity equation

We shall now consider the overall effect when drift, diffusion, generation as well as recombination of carriers occur in a semiconductor. The combined effect is expressed by the continuity equation. Let us consider a bar of semiconductor with cross-sectional area A as shown in Fig. 3.4. Let a slice of infinitesimal thickness (dx) be located at x as shown. The volume of the slice is therefore (Adx). The essence of the **continuity equation** is that the *overall rate of increase in the number of electrons in the given volume is given by the algebraic sum of four components: the rate of flow of electrons into the slice at x minus the rate of flow of electrons out of the slice at x+dx, plus the rate at which electrons are generated in the slice minus the rate at which they recombine in the slice*. Thus the number of electrons in the slice may increase due to the net inflow of

electrons into the slice and/or net generation. Expressed mathematically, this can be written as

$$Adx\frac{\partial n}{\partial t} = J_n(x+dx)\frac{A}{q} - J_n(x)\frac{A}{q} + (G_n - R_n)Adx \text{ .....(3.28)}$$

where G_n and R_n are the generation and recombination rates of electrons. Also, since dx is extremely small, we can write

$$J_n(x+dx) = J_n(x) + \frac{\partial J_n}{\partial x}dx \text{ .....(3.29)}$$

Substituting equation (3.29) into (3.28), we get

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + (G_n - R_n) \text{ .....(3.30a)}$$

Similarly, for the flow of holes

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + (G_p - R_p) \text{ .....(3.30b)}$$

The difference in sign of the current density terms in eqns. (3.30a) and (3.30b) is due to electrons and holes being oppositely charged. Eqns.(3.30a) and (3.30b) are referred to as the **continuity equations** for electrons and holes. Now, substituting the expressions for hole and electron current densities from eqns. 3.21(a) and (b) and the expressions for the net recombination rate as given by eqn.(1.59), the continuity equations for minority carriers (n_p for electrons in p-type semiconductor and p_n for holes in n-type semiconductor) can be expressed as

$$\frac{\partial n_p}{\partial t} = n_p\mu_n\frac{\partial \varepsilon}{\partial x} + \mu_n\varepsilon\frac{\partial n_p}{\partial x} + D_n\frac{\partial^2 n_p}{\partial x^2} + G_n - \frac{n_p - n_{po}}{\tau_n} \text{ .....(3.31a)}$$

$$\frac{\partial p_n}{\partial t} = p_n\mu_p\frac{\partial \varepsilon}{\partial x} + \mu_p\varepsilon\frac{\partial p_n}{\partial x} + D_p\frac{\partial^2 p_n}{\partial x^2} + G_p - \frac{p_n - p_{no}}{\tau_p} \text{ .....(3.31b)}$$

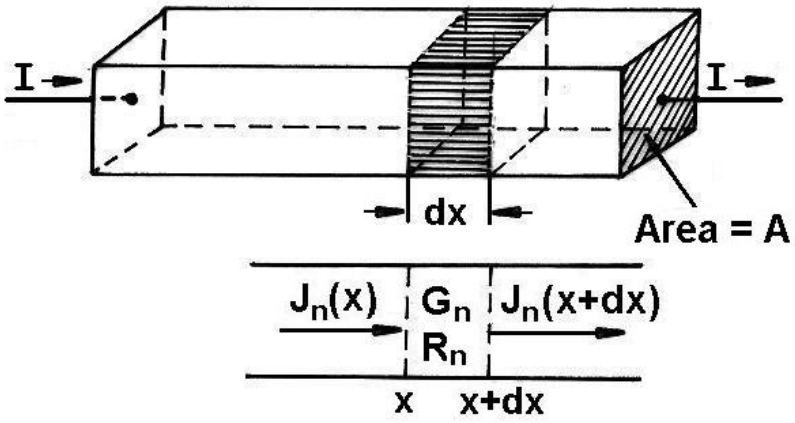


Fig. 3.4: A semiconductor bar with cross-sectional area 'A' showing an infinitesimal slice of thickness 'dx'

Normally, in order to fully understand the operation of a semiconductor device, it is necessary to have complete knowledge of the electron concentration n(x), the hole concentration p(x) and the electric field distribution $\varepsilon(x)$. To obtain a solution for these three quantities, three equations are needed. In principle, the two continuity equations along with the Poisson's equation (to be discussed in Chapter 4) with appropriate boundary conditions should have a unique solution. However, because of the non-linear nature of the equations, analytical solutions are not possible unless simplifying assumptions suitable for each particular case is made. A particular case where the continuity equation will be widely used is the flow of carriers in the absence of electric field, i.e. $\varepsilon = 0$ and the absence of any generation in addition to thermal generation, i.e. $G_n = G_p = 0$. Using these conditions, eqn.(3.31) simplifies to

$$\frac{\partial n_p}{\partial t} = D_n \frac{\partial^2 n_p}{\partial x^2} - \frac{n_p - n_{po}}{\tau_n} \quad \text{.....(3.32a)}$$

$$\frac{\partial p_n}{\partial t} = D_p \frac{\partial^2 p_n}{\partial x^2} - \frac{p_n - p_{no}}{\tau_p} \quad \text{.....(3.32b)}$$

An example of the solution of the carrier transport equation will be discussed next, in order to emphasize the need of suitable assumptions.

3.7 A typical example leading to an expression for Diffusion Length

Let us consider an n-type semiconductor (Fig.3.5) where excess carriers are injected from one side as a result of illumination. This generation of excess carriers increases the carrier concentration at one surface (x = 0) and causes diffusion of the carriers inwards (i.e. into the bulk of the semiconductor). In the steady state condition, the surface generation rate must be balanced by the flux of electrons and holes to the right of x = 0. As the electrons and holes diffuse inside the bulk of the semiconductor, they recombine, resulting in a continuous decrease in their concentrations. Finally, deep in the bulk, the carrier concentrations reduce to their thermal equilibrium values.

Since electrons and holes are generated in pairs, equal number of electrons and holes are injected into the bulk of the semiconductor at x = 0. This implies that the electron current is exactly equal and opposite to the hole current, or in other words, the total current is zero. Also, since these carriers recombine in pairs as they diffuse, even in the bulk $J_p(x) + J_n(x) = 0$. Therefore, from eqns. 3.21(a) and (b), we can write

$$n_n \mu_n \varepsilon + D_n \frac{\partial n'_n}{\partial x} + p_n \mu_p \varepsilon - D_p \frac{\partial p'_n}{\partial x} = 0 \ldots\ldots\ldots (3.33)$$

where n'_n and p'_n are the excess electron and hole concentrations over the thermal equilibrium values. Let us now make a simplifying assumption, called the **quasi-neutrality condition**, in order to obtain a solution of the above equation. This assumption implies that at any time, the excess electron and hole concentrations, as well as their concentration gradients are almost equal at all points in the semiconductor, i.e.

$n_n' \approx p_n'$ and $\frac{\partial n_n'}{\partial x} \approx \frac{\partial p_n'}{\partial x}$. Clearly, the electron and hole concentrations would have been exactly equal if D_n and D_p were equal. However, in practical situations, there is only a small imbalance in electron and hole concentrations, and the quasi-neutrality condition is valid. Using this condition in eqn. (3.33), we have

$$\varepsilon = -\frac{D_n - D_p}{\mu_n n_n + \mu_p p_n}\frac{\partial p_n'}{\partial x} \quad \text{.....(3.34)}$$

Thus, we see that

$$\frac{J_p(\text{drift})}{J_p(\text{diff})} = \frac{\mu_p p_n \varepsilon}{-D_p \frac{\partial p_n'}{\partial x}} = \frac{\mu_p p_n (D_n - D_p)}{(\mu_n n_n + \mu_p p_n) D_p} = \frac{p_n (D_n - D_p)}{D_n n_n + D_p p_n} \quad \text{.....(3.35)}$$

Since $n_n >> p_n$ and $D_n > D_p$, from eqn. (3.35), we can see that

$$\frac{J_p(\text{drift})}{J_p(\text{diff})} << 1$$

In other words, the drift of minority carriers can be neglected, since the internal electric field created due to the charge imbalance is small. However, the drift current for the majority carriers (electrons in this case) is appreciable since it is proportional to the product of the (small) electric field and (large) majority carrier concentration. It is therefore easier to obtain the concentration profiles of the minority carriers analytically, since the possibility of neglecting drift current enables us to simplify the continuity equation.

Neglecting the drift of holes for the n-type semiconductor, we can rewrite the continuity equation (3.31b) as

$$\frac{\partial p_n}{\partial t} = D_p \frac{\partial^2 p_n}{\partial x^2} + G_p - \frac{p_n - p_{no}}{\tau_p} \quad \text{.....(3.36)}$$

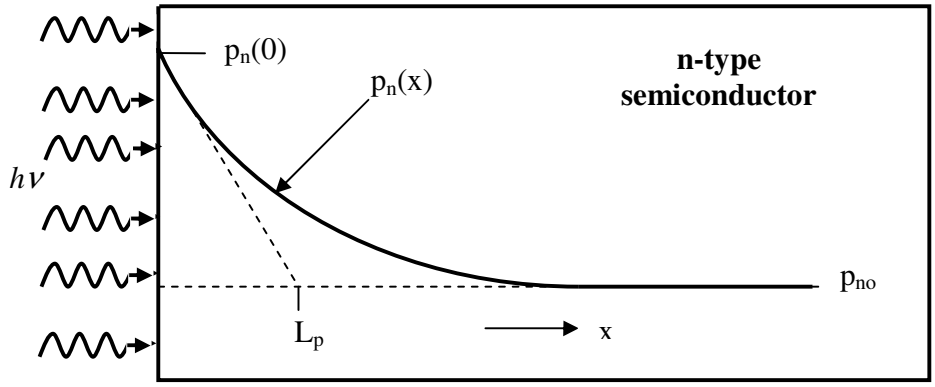


Fig. 3.5: Injection of excess carriers due to illumination from one side of an n-type semiconductor bar and variation of minority carrier concentration as they recombine inside the semiconductor

At steady state, $\partial p_n/\partial t = 0$. Also, $G_p = 0$ towards the right of x = 0 as shown in Fig.3.5(a). Hence eqn. (3.36) can be further simplified to

$$D_p \frac{\partial^2 p_n}{\partial x^2} = \frac{p_n - p_{no}}{\tau_p} \quad \text{.....(3.37)}$$

Noting that $\frac{\partial^2 p_n}{\partial x^2} = \frac{\partial^2 (p_n - p_{no})}{\partial x^2}$, the solution of eqn.(3.37) is of the form

$$p_n - p_{no} = A\exp\left(\frac{x}{\sqrt{D_p \tau_p}}\right) + B\exp\left(-\frac{x}{\sqrt{D_p \tau_p}}\right) \quad \text{.....(3.38)}$$

where A and B are constants to be evaluated from the two boundary conditions given below:

$$\text{At } x=0,\ p_n = p_n(0)$$

$$\text{At } x = \infty,\ p_n = p_{no}$$

where $p_n(0)$ is the concentration of holes at the illuminated surface as shown in Fig. 3.5. Also, as the generated carriers diffuse and move deeper into the bulk they recombine

resulting in a decrease in concentration. At an arbitrary large distance from the surface at which the carriers are generated, the carrier concentration reduces to the thermal equilibrium value. Since p_n is finite, the substitution of these boundary conditions in eqn. (3.38) implies that A = 0 and B = $[p_n(0) - p_{no}]$. Substituting these values of A and B in eqn. (3.38), we can write

$$p_n - p_{no} = (p_n(0) - p_{no})\exp\left(-\frac{x}{L_p}\right) \text{.....(3.39)}$$

where $L_p = \sqrt{D_p \tau_p}$ and is referred to as the **diffusion length**. Substituting eqn. (3.39) into eqn. (3.20) and noting that $p_n - p_{no}$ is actually the excess hole concentration p_n', we obtain an expression for the hole diffusion current in this case as

$$J_{pdiff} = -qD_p \frac{\partial p}{\partial x} = \frac{qD_p}{L_p}(p_n(0) - p_{no})\exp\left(-\frac{x}{L_p}\right) = \frac{qD_p p_n'}{L_p} \text{.....(3.40)}$$

As shown in fig. 3.5, the excess hole concentration dies out exponentially as x increases, due to recombination. The hole diffusion current, which is proportional to the excess hole concentration, also decays at the same rate. L_p represents the distance at which the excess hole concentration falls to $1/e^{th}$ of its value at the point of injection. It should be noted that Fig. 3.5 shows a steady state profile, i.e. the profile does not change with time. This implies that the recombination of carriers in any given region of the profile is compensated by a constant inflow of carriers into the region. The excess inflow, i.e. inflow minus recombination gives the number of carriers which flows out of the region. Thus if we consider any region of the profile between x_1 and x_2 (say), the rate of carriers flowing out at x_2 will be exactly equal to the rate of carriers flowing in at x_1 minus the net

recombination rate between x_1 and x_2. It can also be shown that L_p represents the average distance traveled by a hole before recombination.

Finally it must be emphasized that the assumption of neglecting the minority carrier drift current is not justified in high injection condition. However, as already mentioned in Chapter 1, we shall consider low-level injection in most of the cases to be discussed in this book.

Q3.2. Show that L_p is the average distance traveled by the diffusing carriers before they recombine.

From eqns.(3.39) and (3.40), we see that when carriers are injected from one surface, the steady state carrier density as well as the diffusion current density reduces exponentially with distance. From these equations it can be concluded that if 'I' is the number of carriers which are injected per unit time at x = 0, the number of carriers which have not yet recombined at a distance x is given by $I\exp\left(-\frac{x}{L_p}\right)$. Considering a small slice of thickness dx at a distance x from the injecting surface (as in Fig. 3.4), the number of carriers which have recombined in the slice can be given by

$$I\exp\left(-\frac{x}{L_p}\right)-I\exp\left(-\frac{x+dx}{L_p}\right)=I\exp\left(-\frac{x}{L_p}\right)\left[1-\exp\left(-\frac{dx}{L_p}\right)\right]\approx\frac{Idx}{L_p}\exp\left(-\frac{x}{L_p}\right)$$

Dividing the above expression by I, which is the number of carriers which were injected at x=0, we get the probability of a carrier recombining between x and x+dx as equal to $\frac{dx}{L_p}\exp\left(-\frac{x}{L_p}\right)$. Using the usual method of averaging, the average distance traveled by a carrier before recombining can be expressed as

$$x_{av} = \int_0^{\infty} \frac{x}{L_p} \exp\left(-\frac{x}{L_p}\right) dx = L_p$$

We have thus shown that L_p is the average distance traveled by the carriers before they recombine.

The equations derived in this chapter govern the flow of current in the semiconductor. These, along with Poisson's equation (to be discussed in Chapter 4), control the operation of all semiconductor devices as will be seen in subsequent chapters.